SCRFD-10G
Face Detection Performance Report

DeepStream + TensorRT FP16 · GPU Blur Pipeline

Work Item INTER-69 · Face Detection + GPU Blur

MEASURED REPORT INTER-69-SCRFD-10G · October 2026

Report ID	INTER-69-SCRFD-10G
Work Item	INTER-69 · Face Detection + GPU Blur
Model	SCRFD-10G
Task	Face detection + GPU blur
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Input size	640 × 640
Script	scrfd_face_detection.py
Status	Measured · performance evidence recorded

Prepared in the layout and control style of the xis.ai AI Departmental Handbook v1.0.

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1 Executive Summary

Purpose. This report records measured performance for **SCRFD-10G** in the INTER-69 face detection + GPU blur pipeline on NVIDIA GeForce RTX 5080.

Measured finding. SCRFD-10G runs at 381.4 FPS on a 1920×1080 video file. It detects slightly more faces than 2.5G (45,243 vs 42,802) at moderate GPU cost (30.0%).

1.1 Common settings

Setting	Value
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Model	SCRFD-10G
Detection size	640 × 640
Confidence / NMS	0.35 / 0.4
Frames	1800 (warmup 30 excluded from timings)
Frames dropped	0
Bottleneck	encode (NVENC)

1.2 Summary metrics (video file, 1920×1080)

Metric	Value
Avg Pipeline FPS	381.4
Estimated Max FPS	871.20
Headroom	1.62×
Total Detections	45,243
Avg Faces / Frame	25.14
Max Faces / Frame	37

1.3 Latency per frame (ms)

Stage	mean	max
Detect (nvinfer / TensorRT)	0.68	1.76
Post (decode + NMS)	0.61	1.97
Blur (CUDA kernels)	0.37	0.84
Encode (NVENC)	1.15	2.04

Note: Detect and Blur rows highlighted in green. Total Detections highlighted in yellow.

1.4 Resource usage

Metric	Value
CPU cores used (avg)	1.75
GPU utilisation avg / peak	30.0% / 37%
GPU memory peak	4216 MB

2 Module Execution Placement

Module	Device
Stream Decode (NVDEC)	GPU

Face Detection Model (TensorRT)	GPU
Post-processing (decode + NMS)	CPU (lightweight)
Face Blur (Custom CUDA kernels)	GPU
Video Encode (NVENC)	GPU
Stream Mux / Demux / Queue (DeepStream)	GPU

Almost the entire pipeline runs on GPU. Only the tiny detector output tensor decoding + NMS runs on CPU.

4 Key Takeaways

• **SCRFD-10G** achieved **381.4 FPS** on a 1920×1080 video file with 0 frames dropped.

• Total detections: **45,243** (avg 25.14 faces/frame, max 37).

• GPU utilisation avg / peak: **30.0% / 37%**; memory peak: **4216 MB**.

• Real live streams (cameras / Rutube) are limited by source rate (~25–30 FPS). The pipeline itself is much faster.